PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

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Telkom University
S1 Teknik Elektro (2026)

Mechanical
3D CAD modeling, cycloidal transmission design, and precision fabrication.

Electronics
Schematic circuit design, structured wiring, and power electronics assembly.

Embedded Control
STM32/Arduino MCU firmware, magnetic encoder sensing, and PID/S-Curve control.

Prototyping
Physical build, real-load bench testing, and functional validation workflows.

01 / PROJECTS

Engineering work, in practice.

A collection of 5 projects across undergraduate research, national robotics competition, capstone design, and independent builds.

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01 / FINAL PROJECT



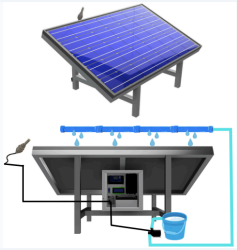
Robot Manipulator 5-DOF ↗

5-DOF articulated robotic arm prototype powered by an STM32F411 microcontroller and 3D-printed cycloidal gearboxes. Integrates AS5600 magnetic encoders in a closed loop to reduce backlash, achieving 79.8% accuracy across 250–1,000 g loads.

STM32F411 MCU · NEMA 17 Stepper · TB6600 Driver · AS5600 Sensor · TCA9548A Mux · Cycloidal Gearbox

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02 / CAPSTONE DESIGN



Pendingin Panel Surya Otomatis ↗

Automated water spray cooling system to lower solar panel surface temperature driven by Arduino Uno. Uses temperature sensing and relay control to trigger a submersible pump above >30°C and shut down below <25°C to restore efficiency.

Arduino Uno R3 · RTD Temp Sensor · 5V Relay Module · 12V Water Pump · 16x2 I2C LCD · 12V Battery Pack

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03 / KRAI (ROBOTICS COMPETITION)



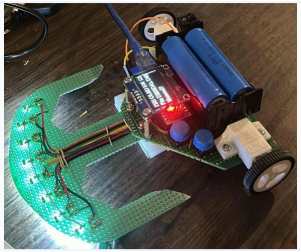
Kompetisi Robot Basket ↗

Mobile basketball robot for Kontes Robot ABU Indonesia (KRAI). Served as a core mechanical team member designing the shooter, dribbler, and drive wheel configuration, advancing to the National Stage at ITB Jatinangor.

6-Bar Pneumatics · 24V Solenoid · RS775 DC Motor · Omniwheel Drive · Limit Switch & IR · Air Regulator

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04 / INDEPENDENT PROJECT



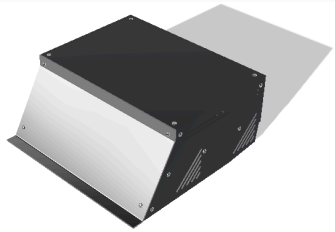
Robot Line Follower ↗

Autonomous line-tracking robot featuring differential drive steering. Equipped with an 8-sensor bar built from LED and photodiode pairs, an Arduino Nano microcontroller, a TB6612FNG driver, and micro N20 metal gear motors.

Photodiode Array · Arduino Nano · TB6612 Driver · N20 Gear Motor · 0.96" I2C OLED · 2S Li-ion Battery

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05 / INDEPENDENT PROJECT



Robot Sumo ↗

1 kg competition-class sumo robot (20 × 20 cm) built on an acrylic chassis with a front aluminum scoop. Delivers 4 kg push force via 4 PG24 planetary gear motors, steered wirelessly from a smartphone via Bluetooth connection.

Arduino Uno · 4x PG24 Motors · L298N Driver · HC-05 Bluetooth · 3S LiPo Battery · Acrylic Chassis

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Telkom University 2026 · Bayu Aji Trisasongko · EIRRG Robotics & Mechatronics Lab

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